

Research Proposal

Title

Improving Real-Time Object Detection Using Transformer-Based Attention Mechanisms

Problem Statement

Current object detection models (YOLO) lack global context understanding. Transformer attention can help the model focus on relevant parts of an image more intelligently.

Objectives

- Integrate self-attention into YOLO backbone
- Reduce false positives in crowded scenes
- Achieve >50 FPS with improved mAP on COCO dataset

Methodology

- Dataset: MS COCO (80 classes, 120k images)
- Baseline: YOLOv8 nano/small
- Modification: Replace conv layers with attention blocks
- Training: 100 epochs, Adam optimizer, lr=0.001

Expected Outcome

- mAP improvement of 2-4% over baseline
- Maintain real-time speed (>30 FPS)
- Published paper / thesis chapter

Timeline

- Month 1-2: Literature review and data preparation
- Month 3-4: Model design and implementation

- Month 5-6: Training, evaluation, and writing